

Diet Manager Version 2.0 – documentation

1. Introduction

The purpose of this project is to design and implement a food and exercise tracking application that enables users to monitor their daily nutritional intake and physical activity.

The system allows users to define foods and recipes, log daily consumption, record exercise, and track calorie balance over time.

The application aims to provide users with a clear overview of their eating habits and energy expenditure. By calculating total calories consumed and burned, as well as displaying nutritional breakdowns, the system helps users make informed decisions about their health and lifestyle.

This document presents the initial design of the system, including its structure, key components, and development approach.

2. Problem Statement and System Overview

2.1 Problem Description

Many individuals struggle to track their daily food intake and exercise consistently. Existing solutions may be overly complex or lack flexibility in defining custom foods and recipes.

Additionally, users often need a way to understand not only calorie intake but also nutritional distribution (fat, carbohydrates, and protein).

Another challenge is combining different types of data:

- Food consumption
- Exercise activity
- Weight tracking
- Calorie goals

These elements must be integrated into a single system that provides meaningful feedback to the user.

2.2 Proposed Solution

The proposed system is a comprehensive tracking application that:

- Allows users to define basic foods and recipes
- Supports nested recipes (recipes within recipes)
- Tracks daily food consumption and exercise
- Calculates:
 - Total calories consumed

- Calories burned through exercise
 - Net calories
- Maintains user weight and calorie goals
- Displays nutritional breakdowns
- Stores all data persistently using CSV files

The system will provide a simple user interface that allows users to input, edit, and view their data efficiently.

3. Team Organization

The project is developed by a team of three members, Denis Keselj, Marko Obsivac, and Gabriel Muskaj. Responsibilities are divided to ensure efficient development and clear ownership of system components.

3.1 Team Roles

- Denis Keselj – Food System, Data Handling, Documentation making
 - Implementation of Food hierarchy (BasicFood, Recipe)
 - CSV parsing and saving (foods.csv)
 - Documentation making and checking activity
 - Making presentation
 - Code handling
- Marko Obsivac – Exercise System and code implementation
 - User interaction and controls
 - Exercise management (exercise.csv)

- Calorie calculations (burned calories, net calories)
 - Gabirel Muskaj– Log System and User Interface vizualization
 - Daily log implementation (log.csv)
 - Implementation of GUI (Swing/JavaFX)
 - Bar chart visualization
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3.2 Collaboration Strategy

- Regular team meetings or correspondention via Whatsapp to synchronize progress
 - Shared repository for version control
 - Clear separation of modules to avoid conflicts
 - Integration testing after each major feature
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4. Development process

This project would be done in numerous phases which will be documented and briefly described. Each of memebbers has its own task in the development process of developing food manager application. Each of the phases would contain main exercises where files, design and java implementation would be done step by step, first showing the prototype of each in version1, and then after the further dealing with the next steps, the final version will be implemented, in order of further testing and finalizing. The next steps are conducted here as follows:

4.1 Phase 1- Initial Planning, Design, and Task Distribution

The first step of the project was focused on understanding the requirements and defining how the system would be organized.

During this stage, the team:

- Discussed the main goals of the application (food tracking, exercise tracking, and calorie balance)
- Defined the core system modules:
 - Food system
 - Exercise system
 - Daily log system
- Designed the basic structure of the application and class relationships
- Created initial documentation draft (this document)
- Distributed responsibilities among team members

Each member received clear tasks to ensure parallel development:

- Denis Keselj – Food system + documentation + CSV for foods
- Marko Obsivac – Exercise system + CSV for exercises
- Gabriel Muskaj – Log system + CSV for daily tracking + GUI implementation

In order to percieve with the further steps, considering java implemmmentation, the team memebbers were also given the code files to implement their exercises:

- Denis Keselj – Food system + documentation + CSV for foods
- Marko Obsivac – Exercise system + CSV for exercises
- Gabriel Muskaj – Log system + CSV for daily tracking + GUI implementation

4.2 Phase 2 – System Design and Diagram Creation

After the initial planning and task distribution were completed, the team continued with designing the internal structure of the application. This phase was very important because it allowed all members to clearly understand how different parts of the system would function together before starting implementation.

The design process focused on creating a logical architecture that would make the system simple to develop, maintain, and later upgrade. Since the application combines several different functionalities, including food tracking, exercise tracking, and daily logging, it was necessary to carefully define the relationships between all classes and data files.

During this phase, the team worked on defining the main class hierarchy of the system. Special attention was given to the food management system, where inheritance was used to separate simple food items from recipes. This approach made the implementation more flexible and allowed support for nested recipes.

In addition to defining the class structure, diagrams were created to visually represent the system. These diagrams provided a clearer overview of the interaction between classes, methods, and CSV files. By using diagrams, the team was able to identify possible structural issues early in the development process and make corrections before implementation began.

This phase served as the technical foundation for the entire project and made the coding process significantly more organized.

The final diagram was made and here it is as follows

Since the project was designed to use CSV files instead of a database, this phase focused on creating well-structured files for persistent storage of all user information.

The decision to use CSV files was made because they provide a simple and lightweight solution for storing data while still allowing easy reading and writing through Java file handling mechanisms.

Each team member was responsible for designing and creating the CSV file related to their assigned module.

Denis Keselj focused on creating the foods.csv file, which stores all food definitions and recipe information. This included defining how nutritional values such as calories, proteins, fats, and carbohydrates would be stored and organized.

Marko Obsivac developed the exercise.csv file, which stores exercise activities together with calorie-burning values. This structure was designed to allow efficient calculation of calories burned during daily activity tracking.

Gabriel Muskaj created the log.csv file, which acts as the main connection point between food intake and exercise tracking. This file records daily user activity and provides the basis for calculating net calorie balance.

During this phase, the team carefully tested different CSV formats to ensure compatibility with future Java parsing methods. This preparation reduced potential implementation problems later in development.

4.4 Phase 4 – Java Implementation

After the planning, design, and data structure preparation were completed, the project entered the main implementation phase. This was the most extensive part of the development process, where all planned concepts were translated into actual Java code.

Implementation was divided according to previously assigned responsibilities, allowing all team members to work simultaneously on separate modules.

Denis Keselj focused on developing the food system. This included implementing the BasicFood and Recipe classes, creating methods for nutritional calculations, and ensuring support for nested recipes. Special attention was required for recursive recipe calculations, since recipes could contain other recipes as ingredients.

Marko Obsivac worked on implementing the exercise system. His work involved creating exercise tracking functionality, implementing calorie burn calculations, and connecting exercise records with daily logs.

Gabriel Muskaj was responsible for implementing the daily log system and graphical user interface. This involved creating user interaction screens, ensuring smooth data entry, and providing a visual representation of user progress.

Throughout the implementation process, regular communication between team members was essential. Since each module depended on interaction with others, frequent testing and integration checks were performed.

The coding phase required several revisions and adjustments as the team discovered opportunities for improving system structure and performance.

4.5 Phase 5 – Testing and Debugging

Once the initial implementation was completed, the team moved to the testing and debugging phase. This stage was necessary to ensure that all modules functioned correctly both individually and when integrated into the final application.

Testing focused on validating all major system functionalities. This included verifying food input and recipe calculations, checking exercise tracking accuracy, testing log generation, and ensuring proper reading and writing of CSV files.

Several issues were discovered during testing. Some of these included incorrect calorie calculations in nested recipes, formatting inconsistencies in CSV outputs, and minor graphical interface alignment problems.

Each identified issue was carefully analyzed and corrected. The team repeated testing after every correction to confirm that the fixes did not introduce new problems into the system.

This iterative debugging process improved overall system stability and reliability.

The testing phase also gave the team an opportunity to evaluate the usability of the application and make improvements to the interface to provide a better user experience.

4.6 Phase 6 – Presentation Preparation

After successful testing and debugging, the final phase focused on preparing the project for presentation and submission.

At this stage, the team reviewed all implemented features to ensure they matched the original project requirements. Additional effort was invested into polishing the codebase, improving readability, and organizing project files.

The documentation was revised and updated to reflect the final state of the project. This included correcting minor inconsistencies and ensuring that all project phases were clearly explained.

The team also prepared presentation materials that summarized the purpose of the project, explained the system design, demonstrated the application functionality, and highlighted the contribution of each team member.

Practice sessions were conducted to ensure that each member could clearly explain their part of the implementation during presentation.

This phase helped transform the project from a technical implementation into a complete professional academic submission.

5. Conclusion

The development of Diet Manager Version 2.0 was a successful team project that achieved its primary objective of creating a functional food and exercise tracking application.

The final system allows users to record daily meals, define custom foods and recipes, log exercise activities, and monitor calorie balance in an organized and practical way. Through nutritional calculations and daily summaries, the application provides useful feedback that can help users better understand their eating and exercise habits.

Beyond the technical outcome, this project provided valuable practical experience in software development.

The team gained deeper understanding of object-oriented programming principles, file handling in Java, modular system design, debugging techniques, and collaborative project development.

One of the most important lessons learned was the value of proper planning and communication. By clearly dividing responsibilities and maintaining regular coordination, the team was able to efficiently combine individual modules into one complete system.

Although the current version of the application successfully meets project requirements, there are several possibilities for future improvement. These include implementing database storage instead of CSV files, adding more advanced graphical analytics, improving user interface design, and potentially developing a mobile version of the application.

Overall, Diet Manager Version 2.0 represents a strong foundation for future expansion and demonstrates successful application of software engineering principles in solving a practical real-world problem.